Math 1040

QUIZ # 6

Name Key

Total 12 points

Please show all your work.

1. (12 pts) The data below are the number of hours 6 students spent online during the weekend and the scores of each student who took a test the following Monday.

a) (8 pts) Find the equation of the regression line.

$y = mx + b$

$m = \frac{n\Sigma(xy) - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$

$b = \bar{y} - m\bar{x}$

	Hours online x	Test score y	xy	x^2
	0	96	0	0
	3	92	276	9
	2	87	174	4
	7	75	525	49
	5	83	415	25
	4	86	344	16
Σ	21	519	1734	103

$\bar{x} = 3.5$ $\bar{y} = 86.5$

$m = \frac{6 \times 1734 - 21 \times 519}{6 \times 103 - 21^2} = \frac{-495}{177}$

$= -2.80$

$b = 86.5 + 2.80 \times 3.5 = 96.3$

Answer $y = -2.80x + 96.3$

b) (2 pts each) Predict the test score if the number of hours spent online is

i) 6

$-2.80 \times 6 + 96.3 = 79.5$

Answer 79.5

ii) 9

$-2.80 \times 9 + 96.3 = 71.1$

Answer 71.1

But as 9 is not in the appropriate range $[0,7]$, this prediction is meaningless